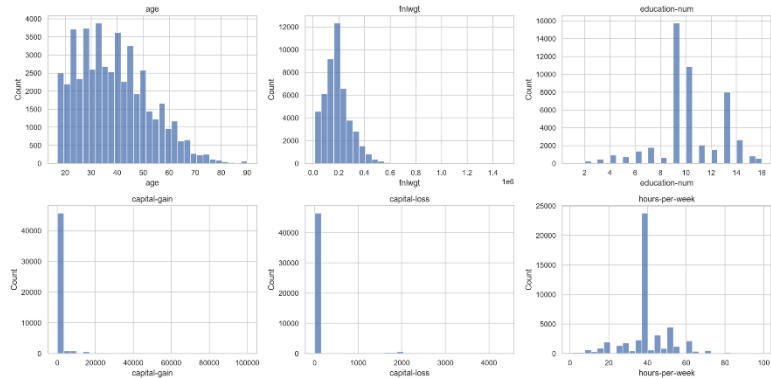


Adult Income Prediction: Problem Framing & Baselines

1. Problem Framing & Metric

Goal: Predict whether a person earns > \$50K/year to prioritize outreach (e.g. for a premium product) toward likely qualified prospects. Positive class: income > 50K. Base rate: 23.9% of individuals earn > 50K (11,687 of 48,842).

Primary metric: F1-score (balances precision and recall), with ROC AUC as a secondary, threshold-free check. We favor precision over pure recall because contacting the wrong person carries a real dollar cost, but F1 keeps recall in the picture so we don't miss too many genuine high-earners.



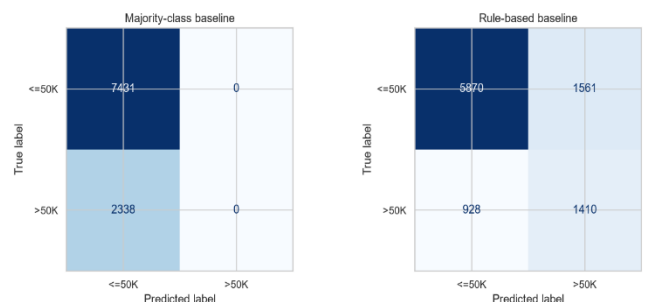
2. Reproducible Split

80% train+dev / 20% hold-out test, stratified on target, random_state=42 (train 34,189 , dev 4,884, test 9,769 rows; ~23.9% positive rate preserved in every split). The hold-out test is touched only once, for final reporting using it during tuning would let the model implicitly fit the test data's quirks, inflating reported performance beyond what it will achieve on genuinely new people.

3. Baseline Results (on hold-out test set)

model	accuracy	precision	recall	f1	roc_auc	pr_auc
Majority-class	0.7607	0	0	0	0.5	0.2393
Rule-based	0.7452	0.4746	0.6031	0.5312	0.7426	0.4946

Majority-class baseline reaches 76.1% accuracy but 0 precision/recall (it never predicts >50K proves that accuracy is misleading here). The rule-based baseline (education-num >= 13 OR capital-gain > 0) trades a little accuracy for real signal: F1=0.531, ROC AUC=0.743. Any real model this week should beat F1=0.531 by a clear margin (+0.05–0.10) to justify its added complexity.



4. Error Analysis & Next Steps

False negatives (missed high earners) skew toward married individuals in exec/professional occupations with high hours-per-week but no capital-gain and sub-Bachelor's education which is a signal our rule ignores. False positives skew toward younger, highly-educated people early in their careers who haven't yet reached > 50K. Missing workclass/occupation values also appear in both error groups. Primary metric going forward: F1-score (secondary: ROC AUC). Baseline to beat: F1 = 0.531, ROC AUC = 0.743 on the hold-out test set.

Fixes:

1. handle missing workclass/occupation/native-country as an explicit "Unknown" category;
2. one-hot/target-encode categoricals;
3. drop redundant string 'education' column (keep education-num);
4. address skew in capital-gain/loss (log-transform or binarize);
5. test with/without fnlwgt (a sampling weight, not a real feature);
6. bucket rare native-country values into "Other".